What a Repair Leaves Behind: Evaluating Meaning-Preservation Interventions for LLM Sentence Expansion in AAC with Benefit, Failure and Cost Together

Anonymous submission

Abstract

Augmentative and alternative communication (AAC) apps increasingly let a large language model expand a few selected words into a sentence spoken on the user's behalf. Fixes for meaning errors in that sentence are usually reported by their targeted gain alone, which hides what they leave unfixed and what they change. We evaluate such interventions offline with a ledger that keeps benefit, *offline failure indicators* (residual failures, edits to already-correct text, displaced readings) and cost side by side, against a cheap comparator and a kill criterion fixed in advance. A runtime guard that regenerates when the output contradicts a cue the user supplied repaired 63/80 scored failures on four LLMs, yet its strict pass rate exceeds that of simply deleting an unsupplied "please" by only 2.08 points [1.15, 3.16]. Frozen and rerun on 78 fresh speech-act item families, it raised joint pass from 78.79% to 88.62% (+9.83 points; deletion alone +3.47), passing every fixed transfer criterion. Yet a pass checks a few labels, not a sentence: one scored repair is "Ask you to wear headphones." A menu intervention shows the same pattern: filling missing readings into three suggestions raised sets offering both polarity or both certainty readings from 15/80 to 78/80, while on other families politeness coverage fell from 26/40 to 12/40. Labels come from an LLM extractor validated against two human annotators on untreated outputs only; repaired text was not human-rated, and we measure no effects on people.

1 Introduction

People who use AAC often compose by selecting a few words or symbols, because every selection costs time and effort (Valencia et al. 2023; Cai et al. 2024). LLM keyword expansion turns `want go party` into a full sentence (Shen et al. 2022; Cai et al. 2024), which is then spoken as the user's own (Hancock, Naaman, and Levy 2020; Valencia et al. 2020). The model chooses more than words: whether something happened or did not, whether the sentence asks or tells, and whether it says "please".

Interventions can reduce wrong choices. A system can check its output against what the user supplied and try again (Madaan et al. 2023; Gou et al. 2024); it can offer suggestions spanning the readings the keywords leave open; or the user can add a cue such as `[NOT]` (Valencia et al. 2024; Mao et al. 2025; Xiang et al. 2025). Such interventions are usually reported by the gain on the endpoint they target. That hides two things when the output speaks for a person: what the fix leaves unfixed, and what it changes that was already right. A repair can restore a request label and leave an incomplete sentence; a menu can gain a missing reading by giving up one the user needed. Offered wording matters: suggested text shifts what people write and believe (Arnold, Chauncey, and Gajos 2020; Jakesch et al. 2023; Williams-Ceci et al. 2026).

We therefore report a ledger rather than a score. Beside each benefit it records *offline failure indicators*: stored-text events that a gain-only report would omit, such as residual failures, edits to already-correct outputs and readings a menu no longer offers. None of them is a measured harm to a person. Our running case is keyword-to-sentence expansion with four LLMs on minimal-pair item families, built in concurrent work that specifies meaning-preservation requirements and tests the scorer itself (Anonymous 2027). Here we ask what each intervention buys, what it leaves, what it costs, and where it applies.

Contributions. (1) An input-conditioned use of benefit–failure–cost accounting: channel-controlled attribution decides what an intervention may be credited with, and paired stored configurations, a zero-call comparator and prespecified kill criteria decide whether it merits a human study (§5). (2) Results led by what a gain-only report would hide: a guard whose increment over a free comparator is modest, whose frozen transfer to fresh items passes, and whose scored passes can be fragments; a menu intervention's collateral loss; and a boundary between ambiguous and supplied input (§6). (3) A replication of a user-side cue on fresh items in which one of our own criteria fails. We ran no study with AAC users; every result is a change in stored text-system behavior on constructed items.

2 Related Work

LLMs in AAC. LLM expansion and suggestion systems for AAC include KWickChat (Shen et al. 2022) and SpeakFaster (Cai et al. 2024), and AAC users have described how such systems help and hinder them (Valencia et al. 2023). Several add user-side control over intent: context and in-

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

tent choices (Valencia et al. 2024; Mao et al. 2025), editable stance tags (Xiang et al. 2025), and retained function words such as "not" (Lacosse, Tano, and McNamee 2026). Others measure whether affect or intent survives generation (Choudhury, Kumar, and Martin 2025; Gorenshtein et al. 2026). Frisch et al. argue that AI-powered AAC needs technical, user-mediated and functional evaluation together (Frisch et al. 2026); our ledger is a technical layer shaped so that a functional study can test it.

Editing, preservation and their evaluation. Generate–critique–revise loops are established (Madaan et al. 2023; Gou et al. 2024), as is conditioning a rewrite on the source dialog act and scoring act retention (Qu, Ungar, and Sedoc 2023). Measuring a targeted change together with collateral change is also established. Ziegenbein et al. train sentence-level edits with rewards for semantic similarity and fluency alongside the target property, so that each edit can be accepted separately (Ziegenbein, Stahl, and Wachsmuth 2026); controlled paraphrasing reports attribute error separately from semantic similarity (Elgaar and Amiri 2025); and style-transfer meta-evaluation shows that content-preservation metrics look valid only when test data do not confound the intended change with content errors (Pauli, Augenstein, and Assent 2025). Detoxifying rewriters default to overly polite output that distorts intent (Wang et al. 2025), and models have default politeness strategies (Zhao and Hawkins 2025). We claim neither the loop nor gain–harm accounting. What we add is accounting conditioned on what the input *supplied*: a checker is credited only with commitments the user gave, a menu only with readings the input left open, and both against a zero-call comparator.

Evaluating AI-mediated communication. Opinionated co-writers shift users' views (Jakesch et al. 2023), AI replies change language and social perception (Hohenstein et al. 2023), and suggestions pull writing toward dominant styles (Agarwal, Naaman, and Vashistha 2025). Those are human-outcome studies of exposure; ours is a pre-deployment study of interventions meant to *prevent* unwanted shifts. LLM judges are common (Zheng et al. 2023) but can agree with each other more than with people on pragmatic labels (Wang, Sun, and Guo 2026; Panickssery, Bowman, and Feng 2024), which bounds what our machine labels can show.

3 Setting

Items and channels. Items are minimal-pair *families*: two source sentences that differ on one critical dimension ("I don't like the new medication" / "I like the new medication") and collapse to the same *bare* keywords (`like medication`). Each source is rendered through five input *channels*: *bare*; *noncrit*, which adds one non-critical word as a length control; *chip1*, which adds the critical cue (`[NOT] like medication`); *chipall*, with cues for all commitments; and *full*, the typed source (`The user typed: ...`). We use three populations from the concurrent study (Anonymous 2027). The *discovery benchmark* has 58 held-out families on six dimensions (polarity, agent, certainty, speech act, politeness, time/quantity), after quarantining two refusal/acceptance families whose scoring rule has a known blind spot; the guard is developed and tested there, and all I2 results use it. *W-C* has 240 fresh families (polarity, speech act, politeness), 238 analyzed after a pre-generation exclusion, with three registered contrasts; its confirmation is qualified because 48 cached requests (12 distinct primary ones) predate the final item freeze. Its 78 speech-act families are the separate population for the frozen guard transfer (§6.2). *W-D* has 80 physical scenarios that add a second cue to a satisfied one. Families are constructed, not messages written by AAC users.

Attribution decides what an intervention may claim. Because the channel fixes which commitments the model received, every miss is attributed by construction (Anonymous 2027). *Loss*: the channel never carried the value. *Violation*: the channel carried it and the output changed it. *Assumption*: the channel was ambiguous and the output committed anyway. A checker can only enforce a commitment the user supplied, so the guard is credited with Violations alone. When the keywords leave a dimension open, no checker can know the intended reading; the remedies are to show alternatives (I2) or to let the user supply the cue (chip).

Systems and scoring. Systems are Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct and Gemma-3-4B-it (4-bit, llama.cpp) and the API model gpt-6.1-terra. P0 asks for one sentence (greedy); P2 asks for three suggestions that "cover the different things the user might mean"; S3 draws three samples at temperature 0.7. An LLM extractor (gpt-6.1-sol, two prompt variants) labels each output's dimensions without seeing source, model or arm; a cross-family arbiter resolves critical-label disagreements, and deterministic code computes verdicts. Treated and untreated outputs therefore pass through the same scoring. In W-C, 259 disagreement rows (177 units) were resolved by a single arbiter after outcomes were known, blind to reference, model and arm; the earlier analysis that masked them as unknown gives the same three conclusions. Against two human annotators on 163 sampled untreated outputs, extractor labels agree at held-out $\kappa = 0.91$–0.97 on all 7 label fields (lowest 95% bound 0.81; pre-specified gate 0.6), and Violation verdicts recomputed from human labels match on 96/99 units ($\kappa = 0.92$). That gate covers extraction on untreated outputs; it does not validate repaired or rewritten text.

4 Interventions

I1: commitment guard (system side, runtime). After the first-pass P0 output, a deterministic lexical monitor (regexes and lexicons for negation, person, modality, question form, numbers, tense and "please") compares the output with the commitments in the visible input: the chips, or for full input the monitor's own labels of the typed sentence. It also flags a politeness marker that was not supplied. On a flag, the same conversation receives one feedback turn ("Your sentence must keep: negated; no added politeness. Write the sentence again with the same meaning otherwise.") and is regenerated, up to two retries. The guard returns the first unflagged attempt, else the attempt with fewest flags, ties going to the original. The monitor never sees the evaluation labels.

G1s (strip-only) is a zero-call comparator that only deletes an unsupplied "please", "pls" or "plz".

I2: ambiguity-complete suggestion set (system side, bare input). Starting from the P2 set, I2 keeps the top suggestion. For polarity and then certainty, if the monitor finds no slot carrying the alternative reading, I2 rewrites the third (then second) slot into that reading with one LLM call ("make it UNCERTAIN: a statement that is not sure (use maybe, might or I think)"), verified by the monitor with one feedback retry. I2 was chosen over a variant that rewrites the top suggestion (I2t) on a 12-family development pilot before the test run.

Chip (user side). The user adds one commitment cue, such as `[NOT]`, `[MAYBE]`, `[?]` or `[PLAIN]`, compared with a length-matched non-critical word at a nominal cost of one selection.

5 Evaluation Design

Paired comparisons of stored configurations. For the system-side interventions, the untreated outcome is the *stored* first-pass output for the same request, and the treated outcome is what the intervention returns from it. This removes between-request variation but does not identify an effect on people, and it does not separate regeneration from serving differences. Pairing matters because greedy decoding is not reproducible: issuing the same request twice reproduced the text in only 76.4–90.8% of local and 44.4–71.4% of API requests. The chip comparison pairs chip1 with noncrit for the same target; these are different requests, so run-to-run churn is part of that contrast.

Estimands and uncertainty. Estimands are row-weighted means over targets or sets; the four models are fixed, not sampled. Discovery intervals are 95% percentile intervals from 2,000 whole-family resamples (seed 20261031); W-C and W-D use 10,000 resamples of recorded clusters or scenarios (seed 20261031); the guard comparator, the transfer and the applicability contrasts use 10,000 whole-family or cluster resamples (seed 20261101). Intervals are pointwise and do not correct for exploratory selection or label error. Thousands of rows are not thousands of independent tests: the guard's 63/80 discovery repairs come from 26 families.

Offline failure indicators and cost. *Changed-correct* counts already-correct outputs whose text the intervention changed, a potential unwanted edit even when *correct→failure*, the share that became scored failures, is zero. *Residual* failures remain after the intervention; *collateral coverage loss* is on dimensions an intervention did not target; *act drift* counts rewrites that change sentence type. We also show what a label cannot see, such as scored passes that are fragments. Cost is extra generator calls and latency; local latency was measured with partial CPU offload and is not a deployment prediction. Kill rules are coarse by design: they decide whether an intervention merits a human study, not whether it is good, and a failing intervention would be reported as killed, not tuned on test.

6 Results

6.1 I1: the guard repairs most scored failures, but beats a free comparator by little on discovery

On discovery cue channels, strict pass rose from 1,312 to 1,375 of 1,392 targets (94.25% to 98.78%; Table 2). Of the 80 initial failures, the guard repaired 63 (78.8% [68.4, 88.9]), far above its 30% kill line. A gain-only report stops there. The zero-call comparator that only deletes an unsupplied "please" already reaches 1,346 passes, so regeneration's increment over it is 2.08 points [1.15, 3.16]: 31 targets pass only with the guard and 2 only with deletion, across 58 families. This compares stored configurations; it does not isolate the contribution of the feedback message.

What remains. 17 of 80 failures remained; 12 involve speech act, and 14/17 were still flagged after both retries: detected, not repaired. For `[?] [YOU] charge wheelchair`, the guard returns the same request twice (Table 2). The guard changed the text of 49/2,154 already-correct outputs across all channels (14/1,312 on cue channels), none into a scored failure; yet "Would you mind repeating the last question" became "Please repeat the last question", still polite by the label and no longer what was typed. It flagged 157/1,856 distinct inputs and added 0.104 generator calls and 0.17 s per input. For typed sentences in an explicitly selected full-sentence mode, returning the input verbatim passes the same checks for 116/116 authored sources at zero calls; this is a baseline, not an expansion method.

A pass is only partial. For `[ASK] [YOU] [PLAIN] wear headphones`, the guard turns "You wear headphones." into "Ask you to wear headphones.", which passes every scored predicate; 4/63 scored repairs begin with this "Ask you" chip echo, a post hoc lexical count rather than a fluency rate. No independent audit of repaired text has been completed, so every repair in this paper is a *scored partial-predicate repair*, not a usable-repair rate. The monitor and the evaluator are separate programs measuring the same constructs (first-pass agreement $\kappa = 0.87$), so a repair is partly a repair as seen by an extractor of the same kind.

6.2 Frozen guard transfer to fresh speech-act items

Speech act was the guard's weakest dimension, so we reran it unchanged (same monitor, feedback text, retry limit and selection rule) on W-C's speech-act families: 78 families, two targets, four models and three cue channels give 1,872 targets (1,848 distinct requests). Transfer criteria were frozen before any retry: at least 30% of strict failures repaired, at most 10% of strict-correct texts changed, and a joint-pass gain whose lower 95% bound exceeds zero. Joint pass rose from 78.79% to 88.62%, 9.83 points [8.39, 11.32] over 78 family clusters. The guard repaired 251/478 strict failures (52.51% [48.36, 56.70]), changed 26/1,394 strict-correct texts (1.87% [1.16, 2.64]) and turned 0 into scored failures; every criterion passed within the call caps (766 of 2,000 generation and 207 of 400 judge-batch requests, failed requests included), with no outcome missing. A separate acceptance audit verified every count, hash and call within the frozen limits; it

	I1 commitment guard	I2 ambiguity-complete set	Chip (user side)
Defect targeted	Violation of a supplied commitment	Assumption under ambiguous input	Loss of a cue at input
Comparison	Stored P0 first pass, same request; strip-only G1s (0 calls)	Stored P2 set, same request; S3 samples	Length-matched non-critical word, same target
Population	Discovery: 1,392 cue targets, 58 families; frozen transfer: W-C speech act, 1,872 targets, 78 families	Discovery: bare-input sets of three, one per family and model	Discovery: 464 targets; W-C speech act: 624 targets, 78 families
Primary benefit	Strict pass; share of scored failures repaired	Sets offering both readings (polarity, certainty)	Critical-value match
Offline failure indicators	Residual failures; correct text changed; correct→failure; scored passes that are fragments	Coverage lost on non-targeted dimensions; rewrites changing sentence type	Matches lost with the chip; misses not repaired; gap to full input
Applicability	Supplied cues only (chips, typed sentence)	Ambiguous input only (§6.4)	Dimension-specific
Cost	Generator calls, latency	Generator calls, latency	Nominal selections
Kill criterion	Repair $\geq$30% of failures *and* change $\leq$10% of correct outputs; transfer adds a joint-pass gain with lower 95% bound > 0	Gain $\geq$30 points over P2	Gain $\geq$10 points over noncrit *and* $\geq$80% of full-input match
Proposed human outcome, horizon	Message sent as meant; corrections; time to send; weeks	Suggestion accepted; meaning endorsed; weeks	Chip use; message sent as meant; weeks

Table 1: Evaluation card: intervention, comparison, population, outcomes and time horizon. All rows except the last are evaluated offline; verdicts are reported with the results (Table 2, Fig. 1). The last row belongs to a proposed, unrun human study (§7). Discovery kill criteria were fixed in the analysis code and checked on development data before the held-out run, without external timestamps; the transfer criteria were frozen in a hashed file before any retry request.

found one untracked readiness-probe request outside the budget ledger, which changes no number. Deletion alone gained 3.47 points, so regeneration's increment over the comparator is larger here than on discovery: 6.36 points [5.29, 7.48].

The residue is larger too: 227/1,872 targets (12.13%) remain strict failures, 215 of them still flagged. Gains concentrate where first passes were worst: 19.02 points for Gemma and 16.24 for Llama, against 3.21 for Qwen and 0.85 for Terra, whose strict repair rates are 18.52% and 35.59%. Cost rises with the failure rate, to 0.41 extra calls per input. These are again scored partial-predicate repairs, and the human extraction gate below covered untreated outputs of the earlier inventory, so it does not validate them. The transfer's declared changed-correct example shows the gap: "Is it okay if I bring a friend?" became "Is [I] bring a friend?", which keeps the question label and so remains a scored pass. The frozen first passes carry W-C's chronology and arbitration qualifications (§8).

6.3 I2: a menu's target gain and its collateral loss

Under bare input, P2 menus offered both polarity or both certainty readings in 15/80 sets; I2 raised this to 78/80, a paired gain of 78.75 points [70.00, 87.50], far above its kill line, at 2.05 extra calls per set (Fig. 1). The slots I2 rewrites are the slots that carried other readings. On separate politeness families, sets offering both a plain and a polite reading fell from 26/40 to 12/40, -35.00 points [-57.50, -15.00]; on speech-act families from 14/31 to 6/31, -25.81 points [-40.62, -9.68]. Conditional on having had both politeness readings, 14/26 sets lost one. Gains and losses are measured on different family strata, not as a within-item trade. In the `knock come` menu (Fig. 2), "Please knock before you come in." becomes "Don't knock before you come in.", removing the polite option. Of 428 rewrites, 138 (32.2%) change sentence type; the instruction asks for "a statement", which a request cannot satisfy without changing type. Coverage is not faithfulness: gained readings can invent details the keywords did not contain.

6.4 Applicability: diversify ambiguous input, preserve supplied input

I2 assumes the input leaves a reading open. W-C also generated menus with the P2 different-meaning prompt for inputs that fix the critical value, which shows what that prompt does there; I2 itself was not tested on supplied input. The most legible endpoint is judge-free: 21/639 full-input polarity menus contain the *other* authored source sentence verbatim after normalization, versus 2/639 sets of three samples. In Fig. 2, "I don't have a doctor's appointment on Tuesday morning." is typed, and the second P2 suggestion is "I have a doctor's appointment on Tuesday morning." The broader categorical endpoint shows the same direction (Table 3): 62.76% of P2 menus versus 2.70% of sample sets contain a focal-label departure, and inspecting only the first suggestion of each still gives 29.14% versus 2.05%. These are not opposite-meaning or user-selection rates, and the comparison does not separate prompt, temperature or joint generation. Its engineering consequence is still direct: diversifying readings that keywords leave open and preserving values the input fixes are different requirements, so diversification, including I2, should be

	Discovery 58 families, 6 dimensions, 1,392 cue targets	**W-C transfer (frozen)** 78 families, speech act, 1,872 targets
Benefit (% of targets)		
Strict pass P0 / G1s / G1	94.25 / 96.70 / 98.78	74.47 / 81.73 / 87.87
Joint pass P0 / G1s / G1	94.68 / 96.77 / 98.85	78.79 / 82.26 / 88.62
G1−P0, points	+4.53 strict	**+9.83** [8.39, 11.32] joint
Strict failures repaired	63/80 (78.8%)	251/478 (52.51%)
Comparator: strip-only G1s, zero calls		
G1s−P0, points	+2.44 strict	+3.47 [2.78, 4.22] joint
G1−G1s, points	**+2.08** [1.15, 3.16] strict	+6.36 [5.29, 7.48] joint
Passes only G1 / only G1s	31 / 2	123 / 4
What remains (offline failure indicators)		
Residual strict failures	17/1,392 (1.2%)	227/1,872 (12.13%)
Correct text changed	14/1,312 cue; 49/2,154 all ch.	26/1,394 (1.87%)
Correct → failure	0/1,312	0/1,394
Cost and verdict		
Extra calls per input	0.104 (1,856 inputs)	0.41 (1,848 inputs)
Kill / transfer rule	passed	passed (all criteria)

Visible input	P0 output	G1 output
Discovery, repaired (Gemma, pol2-14/A, chip1; 1st by hash order; 63/80 scored failures repaired)		
`[PLAIN] let finish sentence`	"Please let me finish my sentence."	"Let me finish the sentence."
Discovery, scored repair that is a fragment (Llama, pol2-07/B, chipall; 2nd by hash order; 4/63 scored repairs begin "Ask you", a post hoc lexical count, not a fluency rate)		
`[ASK] [YOU] [PLAIN] wear headphones`	"You wear headphones."	"Ask you to wear headphones."
Discovery, residual (Llama, act-11/A, chipall; 1st by hash order; 17/80 failures remain, 14/17 still flagged; the source asks whether the listener is charging it)		
`[?] [YOU] charge wheelchair`	"Can you charge my wheelchair?"	"Can you charge my wheelchair?"
W-C transfer, correct text changed, still a scored pass (Qwen, r2act-073/A, chipall; 1st by SHA-256 order; 26/1,394 strict-correct targets changed, none to a scored failure)		
`[?] [I] bring friend`	"Is it okay if I bring a friend?"	"Is [I] bring a friend?"

Table 2: The guard as one display. Left: benefit, the zero-call comparator and what remains, on two *separate* populations that are never pooled. Right: selected stored examples, verbatim. Strict pass keeps the critical value, contradicts no supplied dimension and adds no politeness; joint pass (the transfer's primary endpoint) keeps the critical value and every other scored supplied predicate. Both are scored partial predicates. Brackets: 95% whole-family bootstrap intervals for paired differences. Repair rates condition on initial strict failures; changed-correct and correct→failure on initial strict passes. Examples follow the declared outcome-conditioned rule named in each row and are not typical; keyword inputs omit their stored `Keywords:` wrapper.

Channel	P0 (one)	P2 (three)	S3 (three)	P2−S3, pp
full	2.05	62.76	2.70	60.05 [57.31, 62.70]
chip1	12.09	38.67	12.91	25.76 [23.69, 27.88]
chipall	13.21	37.43	14.46	22.98 [21.38, 24.64]

Table 3: W-C supplied-input channels: % of targets whose output or menu contains at least one focal-label mismatch with the supplied value (full: 1,161/1,850 P2 vs. 50/1,850 S3 complete matched targets; 210 clusters). Exploratory, outside W-C's registered contrasts, in a frozen pre-arbitration snapshot; a categorical mismatch is not necessarily opposite meaning.

conditioned on which dimensions are actually open.

6.5 Chip: replicated on fresh items, still far from full input

On discovery, one chip raised critical-value match from 41.8% [37.1, 45.9] with the length-matched word to 94.8% [91.4, 97.6] (97.3% of full input), passing both kill clauses; speech act was weakest (81.2%). On 78 fresh W-C speech-act families, a registered contrast, the chip raised preservation from 41.83% to 73.24%, 31.41 points [25.00, 37.94] (Holm $p = .0003$), so the gain clause replicates. The $\geq$80% clause does *not*: full input preserved speech act in 99.20%, so one chip reaches 73.8% of it, a paired gap of −25.96 points [−30.36, −21.53]. Question cues fail most visibly: 58/208 question targets became non-questions under chip1, as when `[?] parcel signature`, for a source asking whether a parcel needs a signature, became "Can you sign for this parcel?" (Llama). The nominal cost of one selection shifts the burden of each non-default meaning onto the user (Frisch et al. 2026). A second cue is itself an input intervention: in W-D, Llama under one encoding lost an already-preserved negation in 55/640 transitions, all where `[NOT]` meets `[YOU]` (other cells 0/480), which supports per-model, per-encoding regression tests for cue combinations rather than model-general interference.

6.6 Reading the ledger across interventions

A gain-only report would have credited the guard with a 4.53-point gain where a free deletion explains roughly half of it on discovery, I2 with a 78.75-point gain that displaces readings elsewhere, and the chip with a criterion that fails

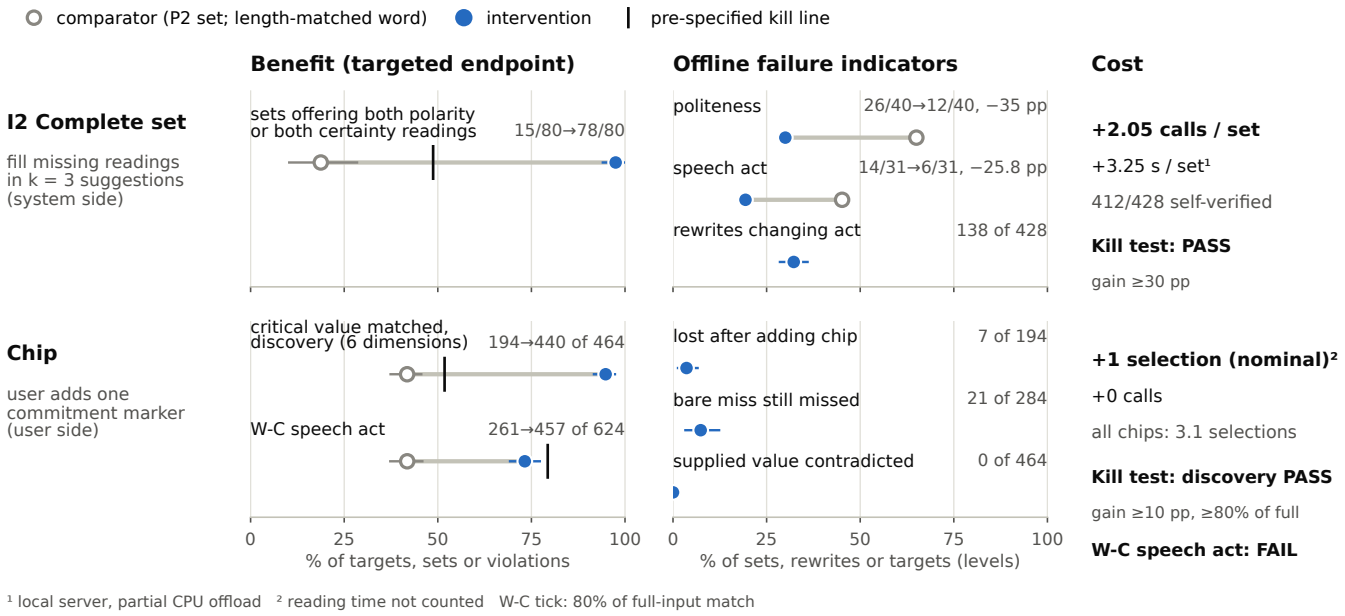


Figure 1: Benefit, offline failure indicators and cost for the menu (I2) and the user-side chip (machine-extracted labels; four models). Open circle: comparator (the P2 set or the length-matched word); filled: intervention; bars: 95% cluster-bootstrap intervals. All plotted values are percent *levels*; percentage-point changes are printed beside them, never plotted. I2's politeness and speech-act rows show the share of matched sets offering both readings, from counts (40 and 31 sets), on families different from its target gain. The chip row adds the W-C speech-act replication, whose tick marks 80% of full-input match. The guard is in Table 2.

on fresh speech-act items. The ledger instead separates what each intervention buys from what it leaves: the guard transfers and helps most where first passes are worst, but its passes are partial; the menu exposes open readings at about twenty times the guard's discovery call cost and should not touch supplied ones; the chip is cheapest per system call and still short of the typed sentence. Effects differ by model, so a deployment decision needs the ledger for the model it ships.

7 A Proposed Human-Outcome Protocol

These are effects on stored text; whether they become effects on people is a causal and longitudinal question. As a proposal only (unrun, not ethics-approved, and requiring design with AAC users as a precondition), a micro-randomized trial (Klasnja et al. 2015) over weeks of users' own keyword-expansion use would randomize only at decision points where the ledger shows the arms differ (e.g., guard output vs. original at the 8.5% of flagged inputs; I2 vs. P2 menus for bare keywords) and estimate causal excursion effects (Boruvka et al. 2018) on proximal outcomes such as sent without edit, corrections and an occasional one-tap "said what I meant", with lagged outcomes for habituation. Chips could only be offered, an encouragement design with its own estimand. If offline changed-correct or coverage loss predicts "said what I meant", the cheap ledger can screen later interventions; if not, evaluation of this kind should be demoted.

8 Limitations

No human outcomes. Nothing here measures communication success, agency, effort or trust; offline failure indicators are not experienced harms. *Machine labels.* Endpoints use an LLM extractor and deterministic verdicts. Its human validation used two annotators on 187 items (a double-coded core with $\alpha = 0.95$–1.00); labels were converted from another format, ties were excluded rather than adjudicated, and agreement is near ceiling on a codebook for explicit markers. It covers untreated outputs only: guard repairs on either population, I2 rewrites and new W-C/W-D boundary cases are not human-validated, and shared monitor/extractor errors can inflate repairs. *Scope of each result.* The guard has two separate populations, the discovery benchmark and a frozen W-C speech-act transfer; I2 was run on neither W-C nor supplied input. W-C's confirmation, and the transfer's frozen first passes, are qualified by early cached requests and by post-outcome arbitration of disagreement labels. The applicability contrasts are exploratory channels outside W-C's registered family, and W-D lacks a complete prospective item/codebook freeze. *Constructed items, fixed models.* Families are authored; four fixed models describe these systems, not AAC users' messages or other models. *Selection.* I2 was chosen over a variant on development data; examples follow declared, outcome-conditioned rules and illustrate rather than sample. Residual and act-drift decompositions were specified after earlier results; their intervals are not corrected for that search.

Menu	Visible input	Configuration 1 (comparator)	Configuration 2	Model, item, declared rule, rate of that rule's population
I2 target gain	`smell gas`	P2: 1. "I smell gas in the kitchen." [plain] 2. "There is gas smell coming from the basement." [plain] 3. "Do you smell gas outside?" [?]	I2: 1. "I smell gas in the kitchen." [plain] 2. "There might be a gas smell coming from the basement."† [hedged] 3. "Do you not smell gas outside?"† [?]	Qwen, cer-11. Rule: first polarity/certainty set gaining both readings (hash order). Sets with both readings 15/80→78/80. Slot 3 also changed and locations were invented.
I2 collateral loss	`knock come`	P2: 1. "Someone knocked and came in." [plain] 2. "Did someone knock before coming in?" [plain] 3. "Please knock before you come in." [polite]	I2: 1. "Someone knocked and came in." [plain] 2. "Maybe someone knocked before coming in."† [plain] 3. "Don't knock before you come in."† [plain]	Terra, pol2-08. Rule: first politeness set losing both readings (hash order). Politeness coverage 26/40→12/40 matched sets; different families from the row above.
Full-input set	`The user typed: I don't have a doctor's appointment on Tuesday morning.`	P2: 1. "I don't have a doctor's appointment on Tuesday morning." 2. "I have a doctor's appointment on Tuesday morning."‡ 3. "Do you have a doctor's appointment on Tuesday morning?"	S3: 1. "I don't have a doctor's appointment on Tuesday morning." 2. "I do not have a doctor's appointment on Tuesday morning." 3. "I don't have a doctor's appointment on Tuesday morning."	Llama, r2pol-024/B, full input. Rule: first full-input P2 set containing the other authored source (‡) (family, model, variant, slot order). Sets reproducing the partner source: P2 21/639 vs. S3 2/639.

Figure 2: Complete three-suggestion menus from two configurations of the same stored request, verbatim, in slot order (keyword inputs shown without their stored `Keywords:` wrapper). Columns compare configurations, not a before and after. Rows follow the outcome-conditioned rule they name (hash order: SHA-1 of request, family, variant and channel), so they illustrate rather than sample; each rate describes that rule's population. Bracketed: the machine-extracted reading ([?] = unclear); †: slots I2 rewrote. The first two rows are discovery bare input; the last is W-C full input, where the hidden source and the visible input coincide.

9 Conclusion

Each intervention passed its pre-specified test, and the frozen guard passed its transfer test on fresh speech-act items. Each also left something a gain-only report omits: residual failures, a modest increment over a free comparator and scored passes that are not sentences for the guard; displaced readings for the completed menu; and, for the user-side cue, preservation far below the typed sentence. Diversification is a remedy for ambiguous input and a source of departures once the input fixes a value. Reporting benefit, offline failure indicators, cost and applicability together turns a ranking question into design questions that a human study with AAC users can test. We release the analysis code, stored outputs and scripts that rebuild every number and figure from them.[PENDING: anonymized artifact URL]

References

Agarwal, D.; Naaman, M.; and Vashistha, A. 2025. AI Suggestions Homogenize Writing Toward Western Styles and Diminish Cultural Nuances. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. ACM.

Anonymous. 2027. Testing What an AAC Expander Is Allowed to Say: Input-Conditioned Commitment Contracts for LLM Sentence Expansion. Concurrent submission to SE4All at ICSE 2027; anonymized copy provided to reviewers.

Arnold, K. C.; Chauncey, K.; and Gajos, K. Z. 2020. Predictive Text Encourages Predictable Writing. In *Proceedings of the 25th International Conference on Intelligent User Interfaces*. ACM.

Boruvka, A.; Almirall, D.; Witkiewitz, K.; and Murphy, S. A. 2018. Assessing Time-Varying Causal Effect Moderation in Mobile Health. *Journal of the American Statistical Association*, 113(523): 1112–1121.

Cai, S.; Venugopalan, S.; Seaver, K.; Xiao, X.; Tomanek, K.; Morris, M. R.; Kane, S.; Narayanan, A.; MacDonald, R. L.; Kornman, E.; Vance, D.; Casey, B.; Gleason, S. M.; Nelson, P. Q.; and Brenner, M. P. 2024. Using large language models to accelerate communication for eye gaze typing users with ALS. *Nature Communications*, 15: 9449.

Choudhury, S. H.; Kumar, A.; and Martin, L. J. 2025. Evaluating Human-LLM Representation Alignment: A Case Study on Affective Sentence Generation for Augmentative and Alternative Communication. In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, 1618–1637. The Asian Federation of Natural Language Processing and The Association for Computational Linguistics.

Elgaar, M.; and Amiri, H. 2025. Linguistically-Controlled Paraphrase Generation. In *Findings of the Association for*

Computational Linguistics: EMNLP 2025, 20842–20864. Association for Computational Linguistics.

Frisch, B.; Wade, W.; Gaines, D.; Kinsella, M.; Peters, B.; Broderick, T.; and Vertanen, K. 2026. It's Complicated: On the Design and Evaluation of AI-Powered AAC Interfaces. Position paper, Speech AI for All workshop at CHI 2026, arXiv:2606.24854.

Gorenshtein, A.; Omar, M.; Jia, E.; Adiniaev, Y.; Daniel, O.; Kruskal, J.; Ahmed, M.; Brook, O.; Klang, E.; and Barash, Y. 2026. Intent Drift in LLM-Assisted BCI Communication: An In-Silico Benchmark Under Simulated Decoder Corruption. MedRxiv preprint, posted 27 August 2026.

Gou, Z.; Shao, Z.; Gong, Y.; Shen, Y.; Yang, Y.; Duan, N.; and Chen, W. 2024. CRITIC: Large Language Models Can Self-Correct with Tool-Interactive Critiquing. In *International Conference on Learning Representations*.

Hancock, J. T.; Naaman, M.; and Levy, K. 2020. AI-Mediated Communication: Definition, Research Agenda, and Ethical Considerations. *Journal of Computer-Mediated Communication*, 25(1): 89–100.

Hohenstein, J.; Kizilcec, R. F.; DiFranzo, D.; Aghajari, Z.; Mieczkowski, H.; Levy, K.; Naaman, M.; Hancock, J.; and Jung, M. F. 2023. Artificial intelligence in communication impacts language and social relationships. *Scientific Reports*, 13: 5487.

Jakesch, M.; Bhat, A.; Buschek, D.; Zalmanson, L.; and Naaman, M. 2023. Co-Writing with Opinionated Language Models Affects Users' Views. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM.

Klasnja, P.; Hekler, E. B.; Shiffman, S.; Boruvka, A.; Almirall, D.; Tewari, A.; and Murphy, S. A. 2015. Microrandomized Trials: An Experimental Design for Developing Just-in-Time Adaptive Interventions. *Health Psychology*, 34(Suppl.): 1220–1228.

Lacosse, E.; Tano, P.; and McNamee, D. C. 2026. Multi-System Cognitive Steering for Generative Communication. Preprint, version 1, posted 13 October 2026; manuscript identifies the BiAlign Workshop at CHI 2026, arXiv:2610.19177.

Madaan, A.; Tandon, N.; Gupta, P.; Hallinan, S.; Gao, L.; Wiegreffe, S.; Alon, U.; Dziri, N.; Prabhumoye, S.; Yang, Y.; Gupta, S.; Majumder, B. P.; Hermann, K.; Welleck, S.; Yazdanbakhsh, A.; and Clark, P. 2023. Self-Refine: Iterative Refinement with Self-Feedback. In *Advances in Neural Information Processing Systems*, volume 36.

Mao, L.; Lee, J. H.; Faroqi-Shah, Y.; and Valencia, S. 2025. Design Probes for AI-Driven AAC: Addressing Complex Communication Needs in Aphasia. In *Proceedings of the 2025 ACM Designing Interactive Systems Conference*, 404–422. ACM.

Panickssery, A.; Bowman, S. R.; and Feng, S. 2024. LLM Evaluators Recognize and Favor Their Own Generations. In *Advances in Neural Information Processing Systems*, volume 37.

Pauli, A. B.; Augenstein, I.; and Assent, I. 2025. Mind the Style Gap: Meta-Evaluation of Style and Attribute Transfer Metrics. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, 21550–21564. Association for Computational Linguistics.

Qu, R.; Ungar, L.; and Sedoc, J. 2023. Conditioning on Dialog Acts improves Empathy Style Transfer. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 13254–13271. Association for Computational Linguistics.

Shen, J.; Yang, B.; Dudley, J.; and Kristensson, P. O. 2022. KWickChat: A Multi-Turn Dialogue System for AAC Using Context-Aware Sentence Generation by Bag-of-Keywords. In *Proceedings of the 27th International Conference on Intelligent User Interfaces*, 853–867. ACM.

Valencia, S.; Cave, R.; Kallarackal, K.; Seaver, K.; Terry, M.; and Kane, S. K. 2023. "The less I type, the better": How AI Language Models can Enhance or Impede Communication for AAC Users. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM.

Valencia, S.; Huynh, J.; Jiang, E. Y.; Wu, Y.; Wan, T.; Zheng, Z.; Admoni, H.; Bigham, J. P.; and Pavel, A. 2024. COMPA: Using Conversation Context to Achieve Common Ground in AAC. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. ACM.

Valencia, S.; Pavel, A.; Santa Maria, J.; Yu, S.; Bigham, J. P.; and Admoni, H. 2020. Conversational Agency in Augmentative and Alternative Communication. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. ACM.

Wang, R.; Sun, K.; and Guo, Y. 2026. Polite but Misaligned: Evaluating LLM Politeness Judgments Against Human Pragmatic Norms. In *Findings of the Association for Computational Linguistics: EMNLP 2026*, 26402–26421. Association for Computational Linguistics.

Wang, X.; Liu, Y.; Pan, J.; Ding, L.; Wang, L.; and Biemann, C. 2025. Chinese Toxic Language Mitigation via Sentiment Polarity Consistent Rewrites. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 35695–35711. Association for Computational Linguistics.

Williams-Ceci, S.; Jakesch, M.; Bhat, A.; Kadoma, K.; Zalmanson, L.; and Naaman, M. 2026. Biased AI writing assistants shift users' attitudes on societal issues. *Science Advances*, 12(11): eadw5578.

Xiang, W.; Xu, Y.; Fang, Y.; Teng, Z.; Jiang, Z.; Hu, B.; and Yang, J. 2025. SocializeChat: A GPT-Based AAC Tool Grounded in Personal Memories to Support Social Communication. Accepted at IEEE SMC 2025 per author record, arXiv:2510.19017.

Zhao, H.; and Hawkins, R. D. 2025. Comparing human and LLM politeness strategies in free production. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 16188–16216. Association for Computational Linguistics.

Zheng, L.; Chiang, W.-L.; Sheng, Y.; Zhuang, S.; Wu, Z.; Zhuang, Y.; Lin, Z.; Li, Z.; Li, D.; Xing, E. P.; Zhang, H.; Gonzalez, J. E.; and Stoica, I. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems*, volume 36.

Ziegenbein, T.; Stahl, M.; and Wachsmuth, H. 2026. Teaching LLMs Human-Like Editing of Inappropriate Argumentation via Reinforcement Learning. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 38616–38637. Association for Computational Linguistics.